

Course Name:	Digital Signal and Image Processing	Semester:	VI
Date of Performance:	<u>18/ 03/ 2026</u>	DIV/ Batch No:	C3
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Introduction

- Mathematical morphology deals with set-theoretic operations. In image processing, morphological operations are typically used to extract information about forms and shapes of structures.
- These are neighbourhood operations which investigate an image using a template image of certain shape and size. This template is called a structuring element.
- The operation can be used to alter the shape (ex. make it bigger or smaller) or detect the presence of a particular form in a given image.

The Structuring Element

The structuring element is a small matrix of pixels akin to the mask used in filtering operations you studied in an earlier experiment. The dimension of this matrix specifies the size of the structuring element and spatial organization of pixel value specifies the shape of the structural element. The size and shape of the structuring element decides the effect of operation on the objects in the image.

Originally morphological operations were defined for binary images and later extended to grayscale images.

Objective: The objective of this experiment is to understand the basics of morphological operations which are used in analyzing the form and shape details of image structures.

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Books Referred:

- A.K.Jain, Fundamentals of Digital Image Processing, Prentice-Hall, 1989.
- Rafael C. Gonzalez, Digital Image Processing, Second Ed., Pearson Education, 2004.
- Bernd Jahne, Digital Image Processing, 5th revised and extended edition, 2002.
- Tinku Acharya, Image Processing Principles and Applications, 2005.
- Dwayne Phillips Image Processing in C, Second Edition, 2000.

Online References:

- <http://www.mathworks.com/help/toolbox/images/fl8-12508.html>
- http://en.wikipedia.org/wiki/Mathematical_morphology

Morphological Operations

Set Theory Basics

Let A and B be sets with elements $a \in A$ and $b \in B$, respectively, and let x be a vector.

Complement

The complement of A is defined as: $A^c = \{d \mid d \notin A\}$

Difference

The difference between two sets A and B is defined as: $D = A - B = \{d \mid d \in A, d \notin B\}$
 $= A \cap B^c$

Translation

The translation of A by x is defined as: $A_x = \{a + x \mid a \in A\}$

Reflection

The reflection of B is defined as: $-B = \{-b \mid b \in B\}$

Dilation

The dilation of X by B is defined as: $X \oplus B = \{x \mid (-B)_x \cap X \neq \emptyset\}$

The result of dilation is a set of all locations x such that $(-B)_x$ has at least one pixel within X.

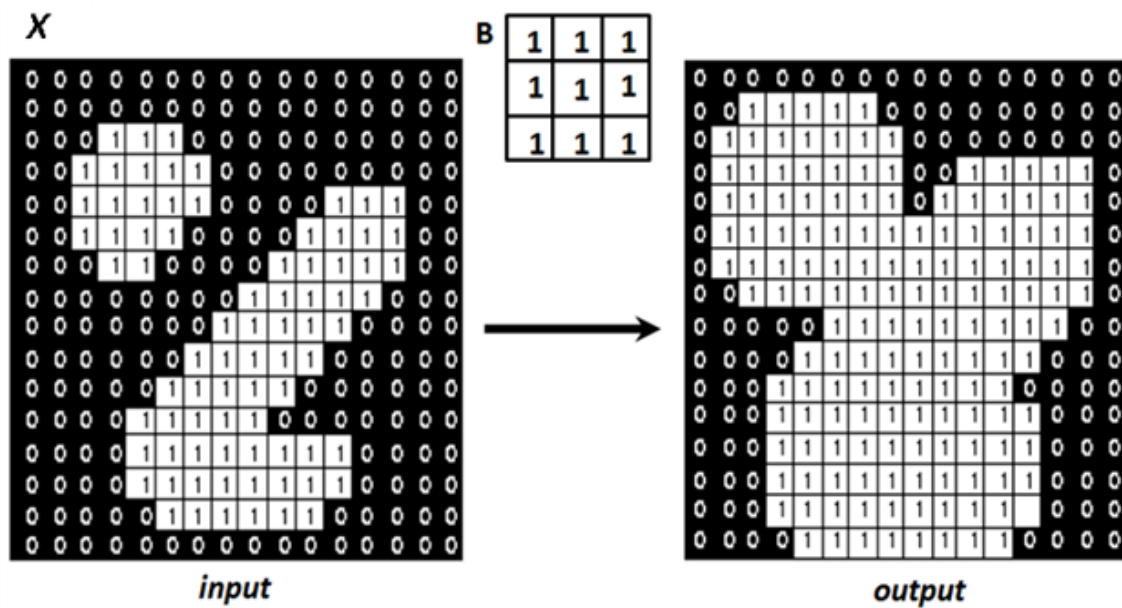


Figure 1. Effect of dilation using a 3X3 square structural element B.

The dilation operation enlarges or expands objects in the image. It connects objects separated by a distance less than the width (window size) of the structuring element. Figure 1 shows the output obtained using a 3×3 square structuring element (whose local origin is at the center of the mask). It can be seen that there is an expansion in object size, and the gap between two different connected components gets filled.

Erosion

The erosion of X by B is defined as: $X \ominus B = \{x \mid B_x \subseteq X\}$

Erosion produces a set of all locations x such that the structuring element B is within (contained in) X .

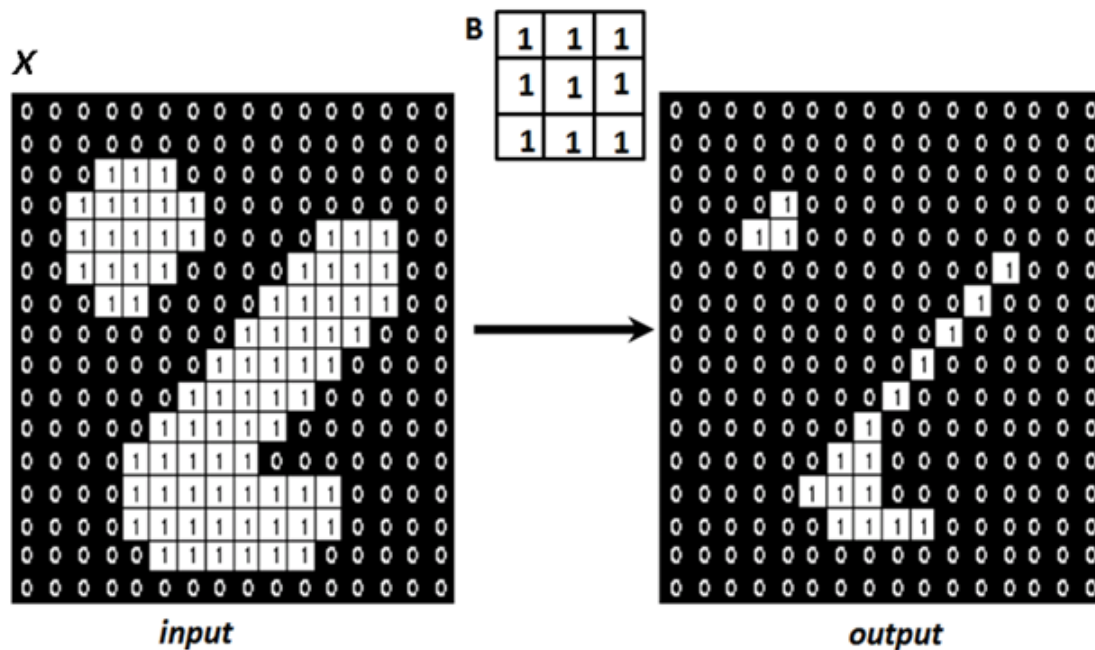


Figure 2. Effect of erosion using a 3X3 square structural element B.

In contrast to dilation, erosion decreases the size of objects. Therefore, the size of the foreground pixels (represented by white pixels) shrinks, and holes within those regions become larger. Figure 2 illustrates results of erosion applied to an example image with the same structuring element.

Opening

By combining erosion and dilation as follows: $X \circ B = (X \ominus B) \oplus B$

We define a new operation called opening. An alternate definition for opening is: $X \circ B = \bigcup \{B_x \mid B_x \subseteq X\}$

$$X \circ B = (X \ominus B) \oplus B$$

The opening operation smooths objects. It breaks narrow bands connecting two large foreground regions and eliminates thin bulges appearing on object boundaries. Figure 3 illustrates such cases.

Closing

By reversing the ordering of erosion and dilation, a closing operation is defined as: $X \bullet B = (X \oplus B) \ominus B$

Closing smooths objects by adding pixels. All smoothing (both for closing and opening) is in relation to the size of the structuring element B .

$$X \bullet B = (X \oplus B) \ominus B$$

The closing operation tends to smooth the section of boundaries but as opposed to opening, it generally fuses narrow breaks and long thin gulfs, eliminates small holes smaller the size of B, and fills gaps in the boundary. Figure 4 highlights results of closing on an example image illustrating effect of closing operation.

Choice of Structuring Element

A key difference between linear filtering, which is also a neighbourhood operation, and morphological processing is that the mask operation is defined in the latter using set theory. The other is the choice of shape. While a square shape dominates linear filtering, the shape of the structuring element is variable and is usually empirically determined based on the application or type of images. Generally, the desirable shape of a structuring element depends on the geometric shape of interest in the image. For instance, a typical medical image contains very few straight lines or objects therefore circular structural element is preferred over oriented structuring element. In contrast, aerial images usually contain oriented objects such as building, roads, etc. therefore oriented or rectangle structuring elements are preferred to better deal with such shapes. The size of the structuring element is determined based on the size of the feature of interest in the image. To preserve larger objects (features) a large structure element is preferred while a small one is used to preserve the finer structural details in the image.

Extension to Grayscale Images

Morphological operations are extendable for gray scale images. In a binary image, the morphological operation is based on set theory. Hence, one can view the 'fit' between B and object X as set intersection.

In grey scale images with pixel values are drawn from 0, 1....L-1 levels. Hence, morphological processing of grey scale images is based on extrema operations. Thus, the output pixel value at any point in a gray-scale image is determined by taking a maximum for dilation or minimum for erosion of the neighborhood pixel values.

Procedure:

The experiment is designed to understand and learn the morphological operations in the images.

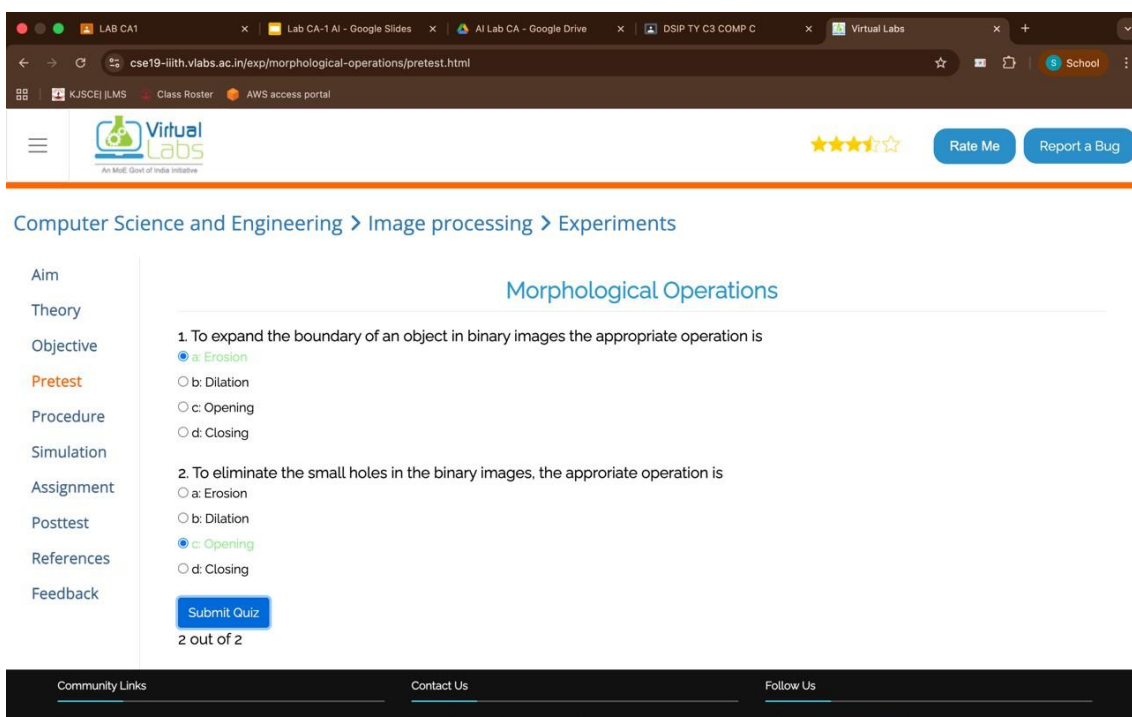
Steps to run the experiments

1. Select image from the mosaic using 'select image' option
2. Select one option from 'Dilation', 'Erosion', 'Closing' and 'Opening'
3. Select options from 'Shape' and 'Size' of structuring elements properties
4. Select run option to perform the operations.

Interesting Observations

1. Select different size of structuring elements and observe the change in final output.

Pretest:



Computer Science and Engineering > Image processing > Experiments

Morphological Operations

Pretest

1. To expand the boundary of an object in binary images the appropriate operation is

☒ a: Erosion

☐ b: Dilation

☐ c: Opening

☐ d: Closing

2. To eliminate the small holes in the binary images, the appropriate operation is

☐ a: Erosion

☐ b: Dilation

☒ c: Opening

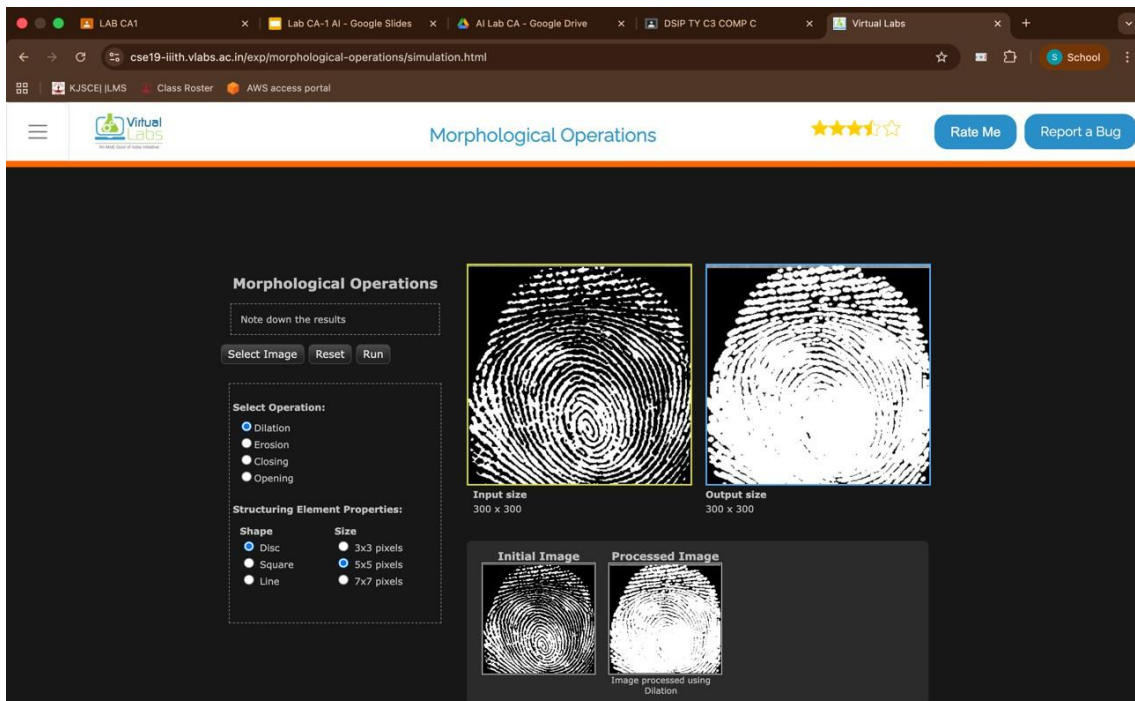
☐ d: Closing

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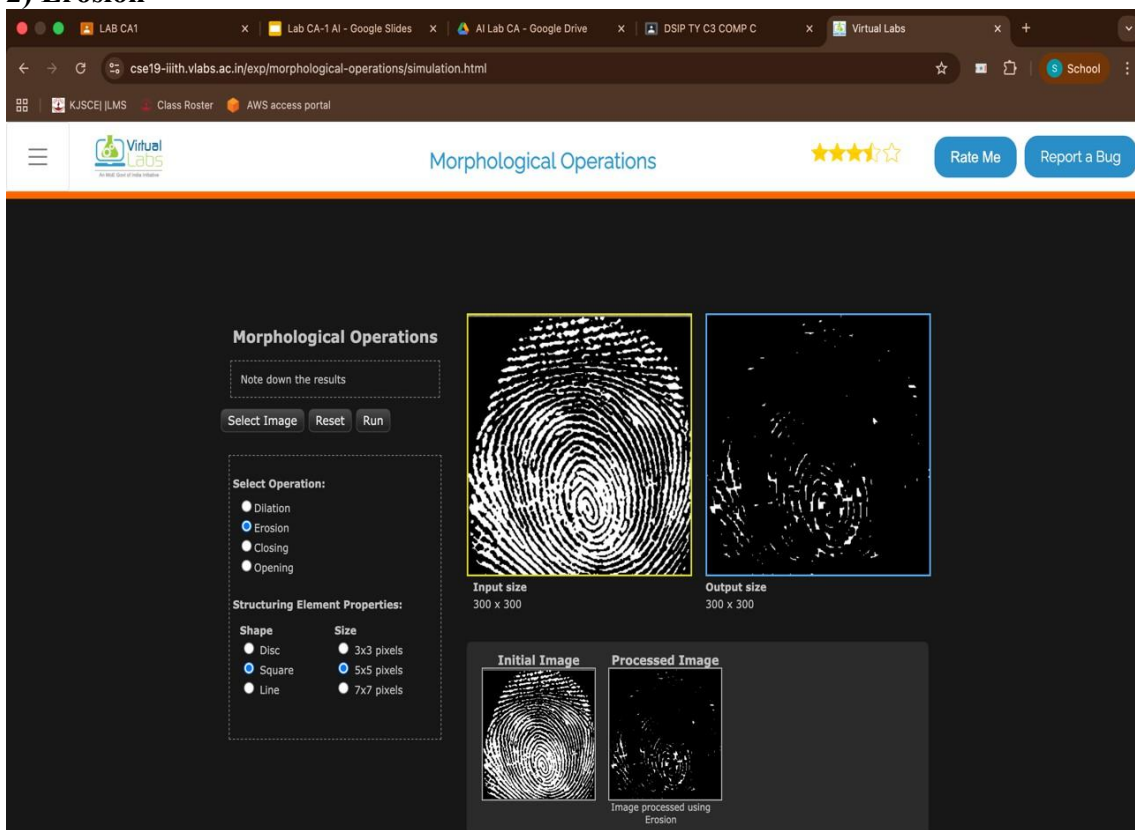
2 out of 2

Simulation:

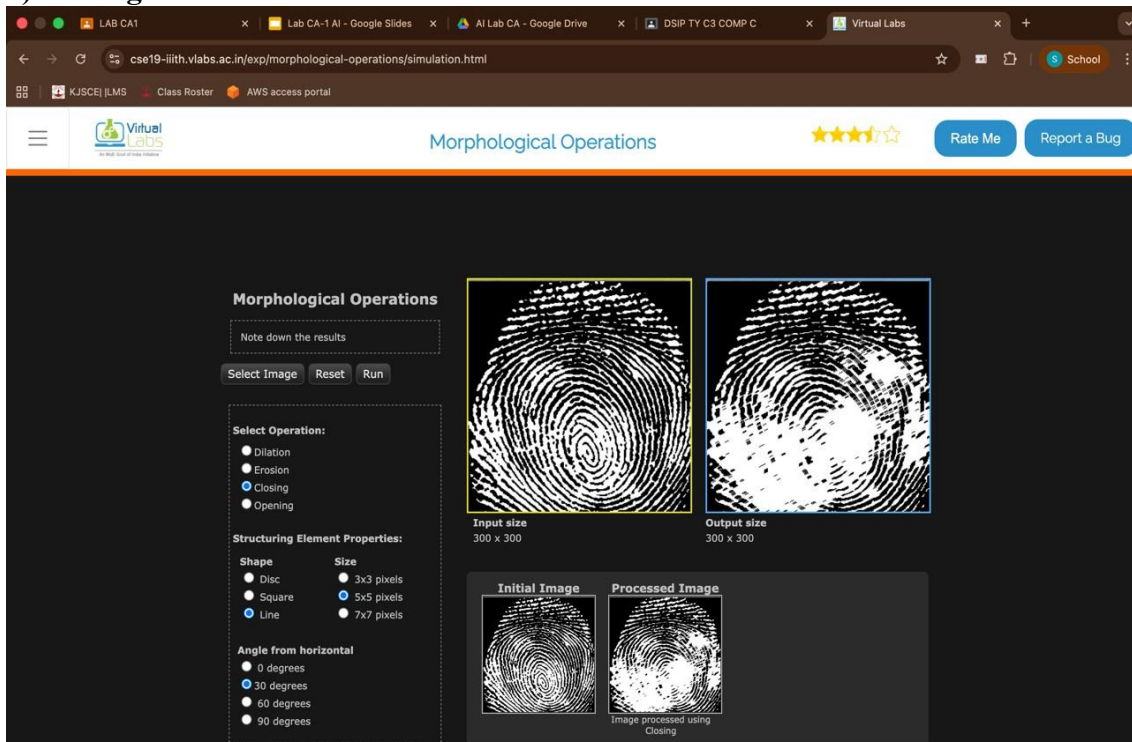
1) Dilation



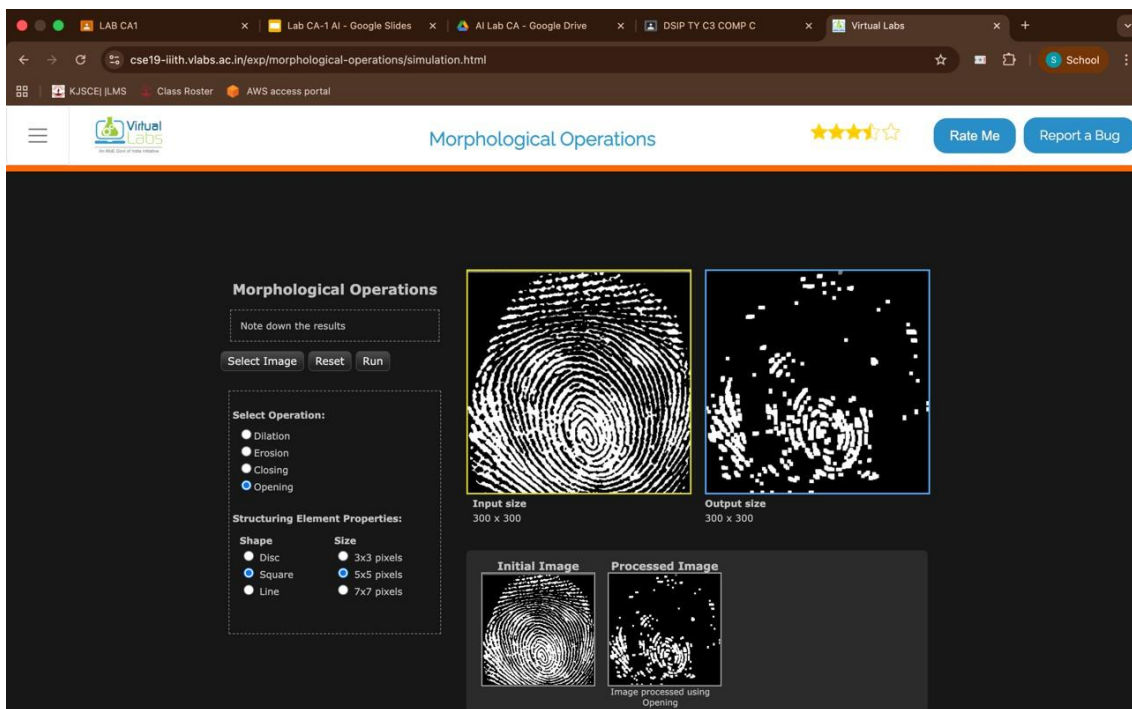
2) Erosion



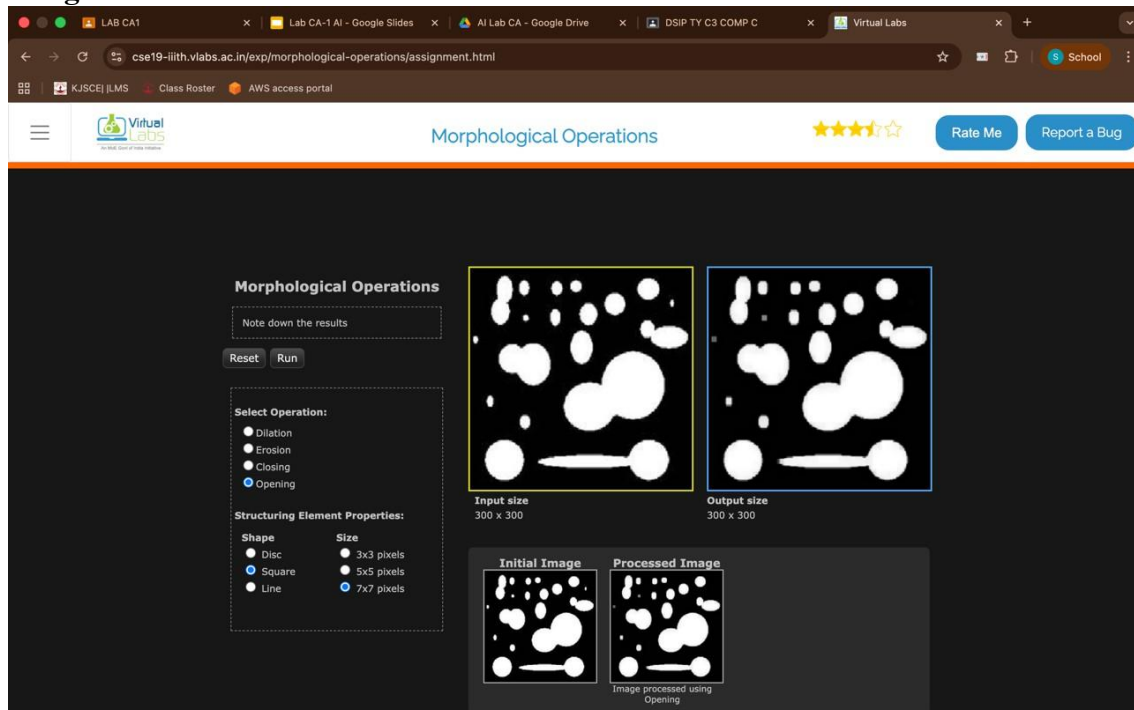
3) Closing



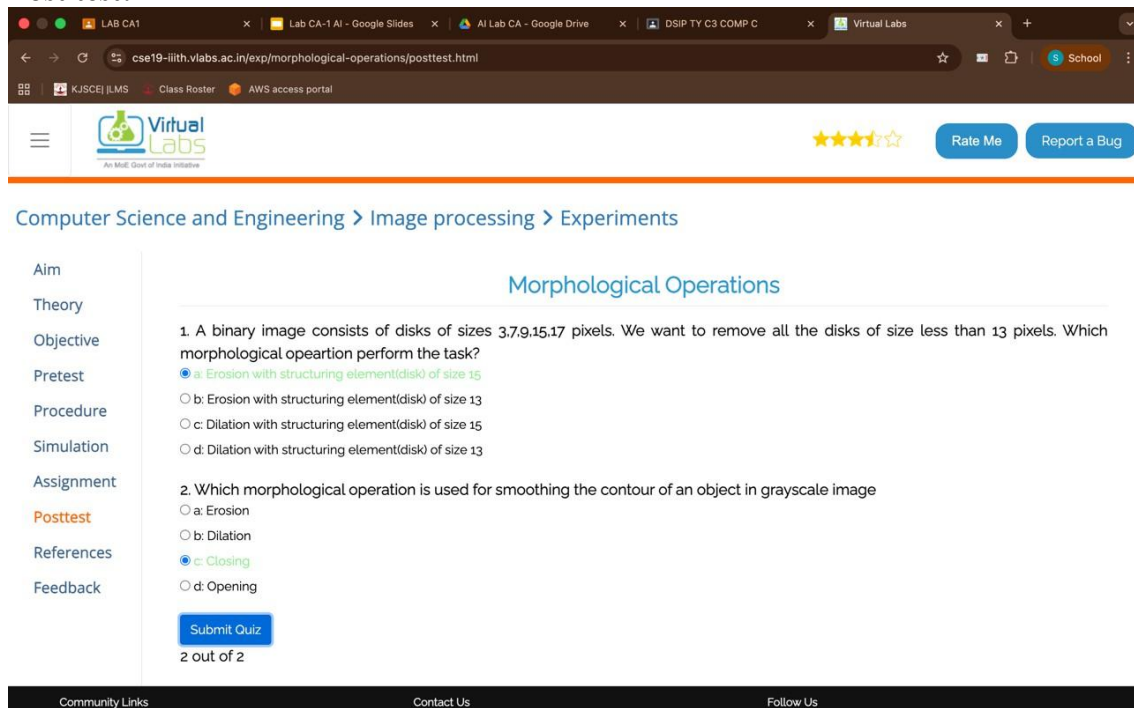
4) Opening



Assignment:



Post test:



Date: _____

Signature of faculty in-charge